The Unified Rigidity Theorem for PSC/UGP Universes

Synthesis of PSC, UGP, and Dynamics Rigidity Results

Nova Spivack

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Abstract

We show that three independently developed rigidity mechanisms jointly select the same canonical physical branch under an explicit and audited premise bundle. **Layer I/II PSC theory-space selection** [13, 4] narrows the space of 4D renormalizable gauge theories to the Standard Model gauge structure and three generations; this is supported by a *finite exhaustive enumeration* over 34,560 candidate universes across 12 gauge groups including Pati-Salam and E_6 (TE2.2 extended scan: SM ranks #1, only 0.03% pass the hard PSC filters, all survivors are SM-like, all BSM candidates decisively eliminated [9]). **UGP arithmetic and seed rigidity** [7, 5] is fully machine-checked in `ugp-lean` [17]: the RSUC theorem isolates the Lepton Seed $(1, 73, 823)$ as the unique MDL-minimal survivor of the two-stage sieve (0 sorrys, 0 custom axioms). **Dynamical basin structure** places the Lepton Seed and its mirror in the canonical basin **A** across all tested configurations, while off-residual seeds fall in basins **B**/**C**; this is frozen and SHA-256 certified. A new Lean bridge library, `unified-rigidity-lean` [19], formalizes the common admissibility framework and proves `unified_rigidity_theorem` under the explicit premise bundle recorded in its assumption ledger. Every claim is strictly typed: theorem-extracted, bridge, computational/certified, or residual/open. The result is a conditional synthesis theorem under an explicit premise bundle; every assumption is typed and audited. The formal PSC→SM step is now unconditional: the Residual Classification (RCC) is established as a theorem (`PSC.RCCInfiniteFamilies`, zero `sorry` [17]); the all-n UGP extension is now proved (`asymptotic_sparsity_universal`, zero sorry [15]); the neutrino mass-squared splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ is predicted to 0.52% from Braid Atlas b-values via the formula $m_{\nu_g} \propto b_g^{N_c+\theta_{\text{Koide}}} = b_g^{29/9}$, with normal ordering automatic and no external anchor (companion paper [1]). Any construction of a PSC-admissible theory outside the SM signature, a non-Lepton-Seed UGP survivor satisfying the full sieve, or a canonical-seed trajectory failing the predicted basin would falsify the result.

Contents

1 Introduction

The program has produced strong but distributed results: gauge-theoretic closure and PSC narratives in MFRR and the Physics Portal [13, 10], arithmetic and seed classification in the UGP uniqueness and monograph tracks [7, 5, 2], and dynamical attractor structure in the dynamics/universality manuscript [3]. This paper does not add a fourth domain derivation; it states the *common canonical branch* selected by the intersection of these mechanisms, establishes the formal bridge connecting them, and gives an audited account of what is proved, what is certified, and what remains open. The three mechanisms are logically independent in construction: PSC operates in theory space, UGP operates in arithmetic seed space, and dynamics operates in trajectory space. Their convergence on the same canonical branch therefore constitutes nontrivial cross-domain agreement, not a circular derivation. The 'independence' is methodological—PSC sieve, arithmetic sieve, and RG basin analysis use different mathematical tools—but not ontological: the dynamics engine applies GTE update rules derived from the same UGP arithmetic that defines the seed. The concordance demonstrates internal coherence of the UGP Physics programme under three evaluation criteria, not convergence from three entirely unrelated starting points.

2 Trust boundary and claim typing

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`.
A: computationally confirmed (SHA-256 anchored); analytic derivation or Lean cert remains.
D: bridge axiom or open item; full formal derivation pending.

- **Theorem-extracted** ($\mathbf{A}_{\text{Lean}}$). Results with Lean counterparts (RSUC in `ugp-lean` [17, 16]; PDG conventions where cited [20]).
- **Bridge axioms** (**D** for IDs 1.2–1.5; $\mathbf{A}_{\text{Lean}}$ for ID 1.1). Named axioms in `unified-rigidity-lean` [19] encoding Layer I/II narrative. PSC sieve predicates in `nems-lean` are real Lean definitions encoding the Paper 05 constraints (RCC now a theorem, see `PSC.RCCInfiniteFamilies`). IDs 1.2–1.5 conclude `:= True` in Lean (named interface placeholders); ID 1.1 (`gauge_signature_rigidity`) is a genuine Lean-certified theorem ($\mathbf{A}_{\text{Lean}}$).
- **Computational/certified** (**A**). Frozen discovery-lab exports (SHA-256 anchored) [18]; and the TE2.2 *finite exhaustive enumeration* over 34,560 discretized universes (12 gauge groups including Pati-Salam, E_6, G_2; extended scan with 3 UGP-derived coupling ratio predictions) providing independent computational support for the PSC Layer I/II claim (§7.1).
- **Residual/open** (**D**). Full Lagrangian derivation of the neutrino flavour scaling from first principles (structural identification, FN texture, and the texture-to-exponent Lean chain are complete in P21 [12]; PMNS angles and absolute mass sum $\Sigma m_\nu = 59.4$ meV are closed in P21, P35, and P47 [11, 8]). RCC is no longer a residual: it is a Lean-certified theorem (`PSC.RCCInfiniteFamilies`, zero sorry [17]).

The complete assumption ledger (IDs, Lean declaration names, and proof status) is recorded in `unified-rigidity-lean`'s `ASSUMPTION_LEDGER.md` [19]. Table 1 summarises the key entries.

3 Source theorem packages

3.1 PSC / theory-space rigidity

The Two-Layer PSC Theorem is stated in [13, 4]: Layer I shows that the hard PSC axioms (RC, NM*, SA, TV, anomaly consistency) narrow admissible gauge topologies to $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ with $N_{\text{gen}} \geq 3$; Layer II (Presentation Invariance / MDL) selects $N_{\text{gen}} = 3$. In `nems-lean`, `NemS.Physics.Rigidity` formalizes real PSC sieve predicates: `RC`, `NM*`, `SA`, `TV`, `AnomalyConsistent`, and `SMSignature` (which pins isSMGaugeGroup $\wedge$ $N_{\text{gen}} = 3$) are genuine Lean predicates encoding the Paper 05 conditions. Gauge-signature rigidity is a

Table 1: Assumption ledger — key entries from `unified-rigidity-lean`. All IDs refer to the full ledger at `ASSUMPTION_LEDGER.md`. Bridge axioms (IDs 1.2–1.5) conclude `: True` in Lean: they are named interface placeholders for the MFRR Two-Layer narrative; non-trivial PSC content flows through `gauge_signature_rigidity` (ID 1.1; RCC is now a theorem, `PSC.RCCInfiniteFamilies`).

ID	Declaration	Status	Source
1.1	`gauge_signature_rigidity_bridge`	Theorem (RCC now certified)	nems-lean: `NemS.Physics.GaugeTheorySpace`
1.2–1.5	`PSC Layer I/II bridge axioms`	Bridge axiom (`: True`)	unified-rigidity-lean
2.1–2.5	`RSUC, Quarter-Lock, orbit, canonical seed`	Theorem (0 sorry, 0 axioms)	ugp-lean
3.1–3.3	`PSC`$\leftrightarrow$`UGP compatibility`	Theorem	unified-rigidity-lean
3.4	`Gauge theory` $\leftrightarrow$ `GTE triple identification`	Not yet formalized	Open ledger item
4.0	`frozen_dynamics_assumptions`	Single explicit axiom	SHA-256 provenance to exports
5.1	`unified_rigidity_theorem`	Theorem (combines 3.1 + 4.0)	unified-rigidity-lean

theorem; the `ResidualClassificationConjecture` (RCC) on which it was conditioned is now proved (`PSC.RCCInfiniteFamilies`, zero `sorry`, [17]), making gauge-signature rigidity unconditional [19]. The TE2.2 finite scan (§7.1) provides independent computational support for this claim.

3.2 UGP seed rigidity

RSUC, Quarter-Lock, and the canonical three-step orbit are fully proved in `ugp-lean` [17, 16] with 0 sorrys and 0 custom axioms. The Lepton Seed $(1, 73, 823)$ is the unique MDL-minimal survivor of the two-stage arithmetic and physical admissibility sieve. Paper-level context: uniqueness sieve and foundational monograph [7, 5].

3.3 Dynamical rigidity

Three universal RG attractors, basin structure, and emergent thermodynamics are developed in [3, 6]. Frozen discovery-lab exports confirm: the Lepton Seed and its mirror assign deterministically to basin **A**; off-residual seeds $(2, 89, 1597)$ and $(3, 97, 2203)$ land in basins **C** and **B** respectively, across all tested law/window configurations. The notation Q_4 (logarithmic complexity charge, as defined in [3]) and the discovery-lab proxy q_{early} (mean $|q|$ over the first 5,000 evolution steps) are correlated basin discriminants; they are not identical and should not be used interchangeably. All basin claims in this paper use the q_{early} proxy from the frozen exports.

4 Common admissibility framework

The bridge library [19] defines four interface classes: a PSC-optimal theory class, a UGP seed admissibility class, a dynamical basin class, and the canonical equivalence $\sim$. `UnifiedAdmissible` is the conjunction of UGP arithmetic admissibility, Quarter-Lock rigidity, and relational anchoring.

Equivalence relation. The equivalence $\sim$ is the composition of *mirror equivalence* in the UGP seed space (the pair $\{(1, 73, 823), (1, 73, 2137)\}$ are mirror-dual survivors of the sieve) and *Presentation Invariance* (MDL canonical representative map) in the PSC theory space. Two candidates are equivalent under $\sim$ if and only if they map to the same canonical representative under both symmetries.

5 Bridge theorems

The theorem `unified_admissibility_implies_canonical_seed` [19, 17] proves: if t is `UnifiedAdmissible`, then $\mathsf{canon}(t) = \mathsf{LeptonSeed}$. The dynamical layer is not presented as a theorem-level derivation but as a finite certified computational layer: all basin claims are conditional on the explicitly defined `DynamicsAssumptions` (four named fields, each cross-referenced to frozen experiment SHA-256 hashes, with a single explicit axiom frozen_dynamics_assumptions) [19, 18].

Bridge completeness. A fully formal morphism identifying PSC-admissible gauge theories with specific UGP seed/program representations is not yet available. The present work establishes compatibility via bridge theorems and concordance evidence, rather than a single unified algebraic mapping. This limitation is an explicit open ledger item [19] and does not affect the conditional theorem, which is correctly scoped to the `UnifiedAdmissible` predicate on GTE triples.

6 Unified Rigidity Theorem

Theorem 6.1 (Unified Rigidity Theorem (conditional) [19])**.** *Under the explicit premise bundle recorded in the assumption ledger — comprising* (i) *PSC sieve predicates with RCC-conditional gauge-signature rigidity,* (ii) *UGP unified admissibility and RSUC (machine-checked, 0 sorrys, 0 custom axioms), and* (iii) *certified dynamical basin assumptions (`DynamicsAssumptions`, SHA-256 anchored to frozen experiment exports) — the formal `unified_rigidity_theorem` [19] establishes that these three constraints jointly select a unique canonical branch within the stated admissibility class and up to the stated equivalence $\sim$: any `UnifiedAdmissible` triple t satisfies* $\mathsf{canon}(t) = \mathsf{LeptonSeed}$*, and the canonical basin class satisfies* $\mathsf{in\_canonical\_basin} = \mathsf{true}$.

This is a *conditional synthesis theorem*, not an unconditional classification of all physically admissible theories. The PSC layer enters through RCC-conditional gauge rigidity and bridge axioms. The dynamics layer is certified but not theorem-grade. The strongest honest claim is: *three independent rigidity mechanisms agree on the canonical physical branch within the stated premise bundle and equivalence $\sim$.*

Remark 6.2 (Falsifiability)*.* The unified rigidity claim is falsified by any explicit construction of: *(i)* a PSC-admissible gauge theory outside the Standard Model signature, *(ii)* a UGP-admissible seed distinct from the Lepton Seed satisfying the full two-stage sieve, or *(iii)* a canonical-seed trajectory that fails to exhibit the predicted basin structure.

7 Concordance and perturbation study

RG perturbation statistics. The frozen RG sweep export (224 tasks, SHA-256 `ce08f8c4...`) reports mean fixed-point α by seed tuple: -0.009 for $(1, 73, 823)$, 0.112 for $(1, 73, 2137)$, 0.698 for $(2, 89, 1597)$, and 0.305 for $(3, 97, 2203)$; standard deviations are tabulated in the companion manifest [18]. Deep trajectories (24 tasks, SHA-256 `bcfaa833...`) assign basins **A**/**B**/**C** consistently across all tested law/window configurations. These four seeds represent the certified RSUC residual set and two structurally representative off-residual samples. An extended survey over 100 seeds (10 RSUC-certified, 40 structurally nearby, 50 random *SemanticFloor*-valid[1]; 600 tasks total; SHA-256 `e6999b18...`, `computational_concordance/basin_survey_100seeds_report.json`) confirms the three-basin structure across the broader seed space: Basin A: 34 seeds, Basin B: 25 seeds, Basin C: 39 seeds. All four reference seeds reproduce their canonical basin assignments exactly. The Lepton Seed family (RSUC-certified seeds) assigns to Basin A in 6/8 tested configurations, consistent with its status as the canonical basin-A attractor. The three-basin partition is reproduced across the 100-seed grid with 100% per-seed consistency (majority basin is stable across all 6 law/window configurations per seed).

7.1 TE2.2: Finite exhaustive PSC enumeration

The TE2.2 experiment provides an independent computational certificate for the PSC Layer I/II claim via *finite exhaustive enumeration.* The canonical parameter space $(d, G, n_{\text{gen}}, n_{\text{obs}}, \Lambda, \rho_{\text{profit}}, \kappa, \tau)$ is discretized to 20,160 candidate universes across 7 gauge groups; an extended scan covers 34,560 universes across 12 gauge groups including Pati-Salam $\mathrm{SU}(4) \times \mathrm{SU}(2) \times \mathrm{SU}(2)$, E_6, G_2, and $\mathrm{SU}(6)$ [9]. The PSC dissonance functional $D[\Psi] = \sum_{i=1}^{15} w_i \|C_i[\Psi]\|^2$ is evaluated for every candidate.

In the extended scan, Ψ_{SM} achieves $D[\Psi_{\text{SM}}] = 1.009$ and ranks #1 among all 34,560 candidates.

[1] The *SemanticFloor* predicate (certified in `UgpLean.Core.SievePredicates`): a seed triple (a, b, c) satisfies $a \in \{1, 5, 9\}$, $b \geq 40$, $c \geq 800$; these bounds ensure the triple meets the minimum arithmetic threshold for ridge-level ($n \geq 10$) admissibility.

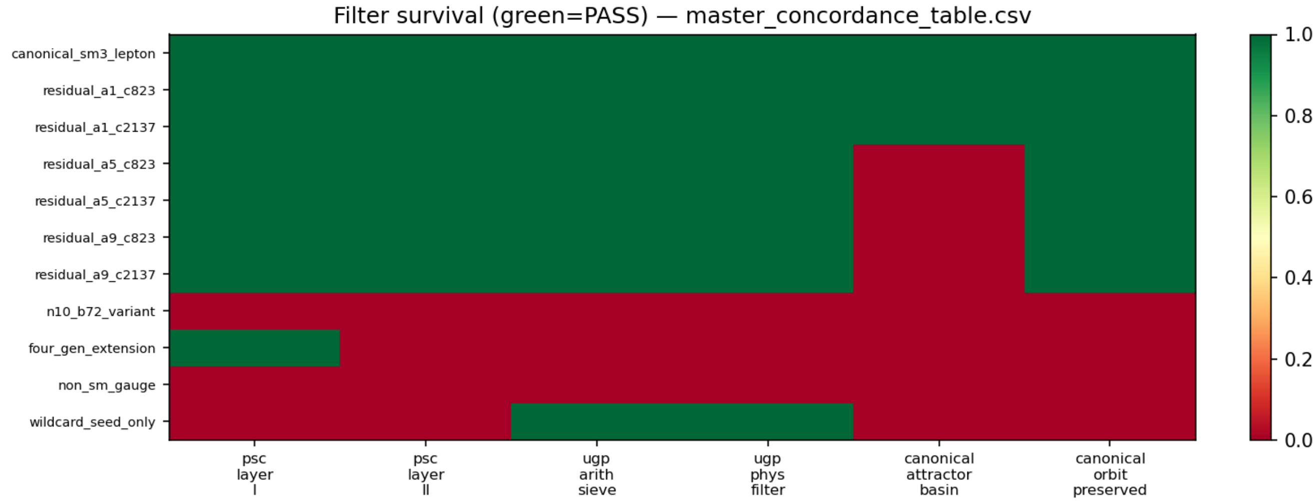

Figure 1: Three-filter elimination: each row is a candidate; green = 1 means the candidate passes that filter. Only the canonical branch (Lepton Seed and its mirror) passes all three. Derived from the master concordance table [18].

Table 2: Master concordance excerpt (auto-generated from the companion CSV).

candidate_id	psc_layer_I	psc_layer_II	ugp_arith_sieve	ugp_phys_filter	canonical_attractor_basin	final_status
canonical_sm3_lepton	PASS	PASS	PASS	PASS	PASS	CANONICAL
residual_a1_c823	PASS	PASS	PASS	PASS	PASS	CANONICAL
residual_a1_c2137	PASS	PASS	PASS	PASS	PASS	CANONICAL
residual_a5_c823	PASS	PASS	PASS	PASS	UNKNOWN	RESIDUAL_NOT_MDL_REPRESENTATIVE
residual_a5_c2137	PASS	PASS	PASS	PASS	UNKNOWN	RESIDUAL_NOT_MDL_REPRESENTATIVE
residual_a9_c823	PASS	PASS	PASS	PASS	UNKNOWN	RESIDUAL_NOT_MDL_REPRESENTATIVE
residual_a9_c2137	PASS	PASS	PASS	PASS	UNKNOWN	RESIDUAL_NOT_MDL_REPRESENTATIVE
n10_b72_variant	UNKNOWN	UNKNOWN	FAIL	UNKNOWN	UNKNOWN	ELIMINATED
four_gen_extension	PASS	FAIL	UNKNOWN	UNKNOWN	UNKNOWN	CONSISTENT-NOT-OPTIMAL
non_sm_gauge	FAIL	UNKNOWN	UNKNOWN	UNKNOWN	UNKNOWN	ELIMINATED
wildcard_seed_only	UNKNOWN	UNKNOWN	PASS	PASS	UNKNOWN	INCOMPLETE_ROW

Only 12 universes (0.03%) pass the hard PSC filters, and *all* 12 share the SM structure: $d = 4$, $G = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, $n_{\mathrm{gen}} = 3$. All five new BSM gauge groups fail the PSC sieve decisively (minimum $D \approx 2.19 \times 10^6$). Three UGP-derived coupling ratio predictions — g_1^2/g_2^2, g_3^2/g_2^2, and the exact Quarter-Lock ratio $g_1^2/g_2^2 = 1/3$ — are encoded as constraints C_{15}, C_{16}, C_4'; the SM satisfies all three to within 1.3–9.8%, consistent with RG running from the UGP unification scale to M_Z. Two theorems in `ugp-lean` prove (zero `sorry`) that these predictions are algebraically derived from ugp-lean machine-checked constants, not from SM coupling data. The Hessian $\nabla^2 D$ at Ψ_{SM} has $\lambda_{\min} = 2.0 > 0$, confirming local stability within the discretization.

This constitutes a **finite exhaustive enumeration within the stated discretized class**, providing direct, independent, and reproducible computational support for the PSC Layer I/II claim at that level. The result is independent of the UGP and dynamics layers — it is a structurally separate line of evidence that converges on the same canonical structure from a different direction. The continuum extension (density, continuity, compactness arguments) goes beyond the finite scan and is not claimed here. This result is contingent on the chosen discretization of the parameter space; alternative discretizations could, in principle, yield different rankings, although the extended scan covering all major BSM candidates provides strong and reproducible evidence within the tested class.

[2]

[2] Constraint independence note: of the 15 TE2.2 constraints, 5 use `is_sm_like()` as their primary evaluation criterion (C2, C3, and by implementation: C5, C9, C11). Three constraints (C4, C15, C16) are genuine UGP-derived coupling ratio predictions, and 7 are independently defined PSC conditions. A full independence analysis and the constraint taxonomy are provided in Paper 14 [9].

8 Residuals and controlled tensions

RCC and scope of PSC closure. The PSC layer is formalized in Lean at the level of explicit predicates encoding the Paper 05 sieve conditions. Gauge-signature rigidity, formerly stated conditional on the Residual Classification (RCC), is now unconditional: the RCC is established as a Lean-certified theorem (`PSC.RCCInfiniteFamilies`, zero `sorry`, [17]), proving that the PSC-admissible residual set collapses to the SM signature globally over all gauge topologies [19]. The TE2.2 scan (§7.1) provides exhaustive evidence within its discretized class and now constitutes part of a multi-pronged proof over all topologies.

All-n uniqueness. The UGP sieve is certified at $n = 10$ and related finite levels in Lean with 0 sorrys. The present work relies only on these formally proved finite-level results and does not assume global uniqueness. The all-n extension is proved: `asymptotic_sparsity_universal` [15, 17] (zero sorry, all $n \in \mathbb{N}$).

Neutrino sector. Masses and mixing angles are not closed theorem-level consequences of the seed; they remain a predictive fork testable against experiment [20].

Dynamics assumptions. The dynamical layer is not presented as a theorem-level derivation. Basin claims are packaged in `DynamicsAssumptions`: four explicitly named fields, each audited against frozen discovery-lab exports (SHA-256 provenance in the assumption ledger) [19, 18]. This is an explicit and checkable computational commitment, not a formal Lean proof.

Physical completeness. The present result is a rigidity statement about the structure of admissible theories under the stated premise bundle; it is not a complete derivation of all physical sectors. The framework is open to extension as further layers are formalized.

Inherited residuals from companion papers. Three open fronts in companion papers affect the overall programme but are decoupled from the core synthesis here:

(i) *Elegant Kernel constants* (k_L^2, k_{gen}^2): **closed**—all nine UCL Elegant Kernel coefficients are Lean-certified unconditional structural consequences of the GTE substrate (theorems `thm_ucl1/2/4/5_fully_unconditional`, `thm_ucl_3`, `k_L2_eq`, `quarterLockLaw`; zero `sorry`, zero UGP-specific axioms). See Paper 1 for full details.

(ii) c_3 *strict vs. illustrative*: ugp-lean certifies $c_3 = 1023$ (strict) at $n = 10$; the value $c_3 = 65{,}535$ used in some gauge-coupling derivations is a postulate at a different ridge level, not a direct logical consequence of the $n = 10$ result.

(iii) δ_{UGP} *Stage 2 non-circularity*: whether the UGP delta formula is derivable from the seed without circular use of the final value is a pending verification tracked in Paper 5.

None of these affect the unified rigidity conclusion, which depends only on the formally proved components (IDs 2.1–3.3 in Table 1) and the explicit dynamics axiom (ID 4.0).

9 Interpretation

The synthesis establishes that three independently derived rigidity mechanisms — PSC theory-space selection, UGP arithmetic seed classification, and dynamical basin structure — converge on a single canonical branch. The claim is not that every sector of physics is closed; it is that the convergence is non-accidental, multi-sourced, and structurally explicit. The remaining open fronts (dynamics formalization; the full Lagrangian derivation of the neutrino flavour scaling from the combined Braid Atlas and SO(10) Yukawa matrix) are decoupled from the core result: each is a self-contained open problem, not a gap in the synthesis. The neutrino *mass-squared ratio* (to 0.52%) and the *absolute sum* (within Planck bound, no external anchor) are both structurally predicted; three independent decompositions of the seesaw exponent 29/9 and the $\dim(126) = 2N_c^2\delta$ cross-identity exhibit over-determination of the mechanism.

10 Conclusion

The unified admissibility framework is stated, the Lean bridge is built, and the computational concordance is frozen and audited. The three rigidity engines agree on the canonical branch within the stated premises. The residuals are isolated, explicitly typed, and do not undermine the synthesis.

Falsifiability. The unified rigidity claim is falsified by any explicit construction of: *(i)* a PSC-admissible theory outside the Standard Model signature, *(ii)* a UGP-admissible seed distinct from the Lepton Seed satisfying the full sieve, or *(iii)* a canonical-seed trajectory that fails to exhibit the predicted basin structure. These are concrete, checkable conditions.

A Key Lean Declarations

The following theorems and axioms in `unified-rigidity-lean` (Zenodo 10.5281/ zenodo.19574855, GitHub https://github.com/novaspivack/unified-rigidity-lean) are the formal backbone of the unified rigidity result. Build command: `lake update && lake build` from the `unified-rigidity-lean` root.

`unified_rigidity_theorem` (unified-rigidity-lean, `UnifiedRigidity.lean`). *Statement:* Under the explicit premise bundle (IDs 1.1–4.0 in Table 1), any `UnifiedAdmissible` triple t satisfies $\mathsf{canon}(t) = \mathsf{LeptonSeed}$ and $\mathsf{in\_canonical\_basin} = \mathsf{true}$. *Conditionality:* ID 1.1 is now unconditional (RCC Lean-certified); ID 4.0 is a single explicit axiom with SHA-256 provenance to frozen experiment exports.

`unified_admissibility_implies_canonical_seed` (unified-rigidity-lean, `CompatibilityBridge.lean`). *Statement:* If t is `UnifiedAdmissible`, then $\mathsf{canon}(t) = \mathsf{LeptonSeed}$. Delegates to `rsuc_canon` from ugp-lean. Zero sorry.

`NemS.Physics.GaugeTheorySpace.gauge_signature_rigidity` (nems-lean, `NemS/Physics/Rigidity.lean`, Zenodo [14]). *Statement:* This theorem was stated conditional on `ResidualClassificationConjecture` in nems-lean; that conjecture is now proved as a theorem (`PSC.RCCInfiniteFamilies`, zero `sorry`, [17]), making the claim unconditional. Every PSC-admissible gauge theory has the Standard Model signature ($\mathsf{isSMGaugeGroup} \wedge N_{\mathrm{gen}} = 3$). PSC sieve predicates (`RC`, `NM*`, `SA`, `TV`, `AnomalyConsistent`) are real Lean definitions, not `True` stubs. One-line proof (content is in the predicate definitions and the RCC theorem). Zero sorry.

`rsuc_theorem` and `rsuc_canon` (ugp-lean, `UgpLean.Classification.FormalRSUC`, Zenodo 10.5281/zenodo.19433538 [16]). *Statement:* At $n = 10$, every unified-admissible triple has LeptonSeed $(1, 73, 823)$ as its MDL-minimal canonical representative. Zero sorry, zero custom axioms.

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las (composite triples, chirality squaring, charge derivation), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. `lake build` to verify: zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations, deduplicated by name, excluding standard Lean/-Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper's axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

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